E-01 COVER AND CONTROLLED REVISION STATUS

FC01 Compact Filling-Cell Electrical Design Package - Revision E

Selected architecture: CPU 1511-1 PN, local S7-1500 I/O, MTP700 Unified, two G120C PN drives and non-safety NVIDIA inspection.

Native source generated from the controlled Revision-D.1 schedules and first validated with QElectroTech 0.100.0.

OPEN: site supply, fault current, motor data, cable routes, ambient, enclosure duty and qualified safety engineering.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

E-02 DRAWING INDEX

REV E | 2/26

E-01 Cover and controlled revision status

E-02 Drawing index

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E-06 24 VDC PELV distribution

E-07 S7-1500 rack and module allocation

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E-14 Conveyor G120C power and PROFINET control

E-15 Pump G120C power and PROFINET control

E-16 Motor cables, PE and shield termination

E-17 PROFINET and OT network architecture

E-18 HMI and NVIDIA inspection interface

E-19 Solenoid valves and interposing relays

E-20 Field sensors, flow and level instrumentation

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E-22 Terminal strips X100 and X101

E-23 Terminal strips X102/X103, power, common and shields

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E-25 Conceptual safety interface and boundaries

E-26 Spare capacity, BOM coverage and open inputs

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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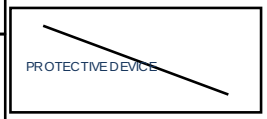
Author: FC01 engineering team	Drawing index	File: FC01_revision_e.qet
Date:		Folio: 2/26

E-03 DESIGN BASIS, LEGEND AND REFERENCE DESIGNATIONS

REV E | 3/26



=FC01+CP01-QS100

Main isolation

=FC01+CP01-A100

PLC/controller

=FC01+FLD-FT100

Field instrument

=FC01+FLD-YV100

Solenoid actuator

=FC01+CP01-X100..-X199

Terminal family

Designation form: =FC01+<location>-<tag>; CP01 control panel, FD01 field, OP01 operator and IF01 external capper interface.

Ordinary PNP 24 VDC field sensing; 4-20 mA analog; TM Count pulse channels; PROFINET Standard Telegram 1 for drives.

PROVISIONAL: 400/230 VAC 3P+N+PE 50 Hz TN-S and 24 VDC PELV are design assumptions pending site confirmation.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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E-04 SINGLE-LINE INCOMING POWER DISTRIBUTION

REV E | 4/26

=FC01+FLD-GL100
Site cable entry
3P+N+PE assumption

=FC01+CP01-QS100
Door-coupled main isolator

=FC01+CP01-QF100
Main protective device

=FC01+CP01-PE100
PE bar and enclosure bonding

=FC01+CP01-X0
Incoming distribution terminals

Functional single-line path only; conductor sizing and protection ratings remain open and are not construction values.

PE is not switched. Bond enclosure, mounting plate, door and equipment PE to PE100 using verified conductors.

OPEN: supply earthing system, prospective fault current, SCCR, selectivity, voltage drop and installation method.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Single-line incoming power distribution	File: FC01_revision_e.qet
Date:		Folio: 4/26

E-05 MAIN DISTRIBUTION AND BRANCH PROTECTION

REV E | 5/26

PROTECTIVE DEVICE

=FC01+CP01-QF100

Main protective device

VARIABLE-FREQUENCY DRIVE

=FC01+CP01-U100

Conveyor G120C PN

VARIABLE-FREQUENCY DRIVE

=FC01+CP01-U101

Pump G120C PN

POWER SUPPLY

=FC01+CP01-PS100

24 VDC 20 A supply

PROTECTIVE DEVICE

=FC01+CP01-QF110

24 VDC branch protection

Separate mains/drive routes from 24 VDC control, instrumentation and Ethernet wiring.

Drive input protection, EMC accessories, braking duty and motor cable selections are provisional pending nameplates and vendor coordination.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Author: FC01 engineering team	Main distribution and branch protection	File: FC01_revision_e.qet
Date:		Folio: 5/26

E-06 24 VDC PELV DISTRIBUTION

REV E | 6/26

POWER SUPPLY

=FC01+CP01-PS100

24 VDC 20 A supply

PROTECTIVE DEVICE

=FC01+CP01-QF110

Electronic branch protection

PLC / MODULE

=FC01+CP01-A100

PLC and local modules

NETWORK NODE

=FC01+CP01-SW100

Managed switch

NETWORK NODE

=FC01+CP01-PC200

NVIDIA edge compute

Protected branches: PLC/I-O, HMI, relays/solenoids, network/security and vision/lighting; exact channel ratings remain a native design input.

0 V reference and functional-earth/shield paths are controlled separately; do not substitute shield drain for protective earth.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Author: FC01 engineering team	24 VDC PELV distribution	File: FC01_revision_e.qet
Date:		Folio: 6/26

E-07 S7-1500 RACK AND MODULE ALLOCATION

REV E | 7/26

PLC / MODULE

=FC01+CP01-A100
SIMATIC S7-1500 CPU 1511-1 PN
6ES7511-1AL03-0AB0

PLC / MODULE

=FC01+CP01-A101
TM Count 2x24V
6ES7550-1AA01-0AB0

PLC / MODULE

=FC01+CP01-A102
DI 32x24V DC HF
6ES7521-1BL00-0AB0

PLC / MODULE

=FC01+CP01-A103
DQ 32x24V DC/0.5A HF
6ES7522-1BL01-0AB0

PLC / MODULE

=FC01+CP01-A104
AI 8xU/I HF
6ES7531-7NF00-0AB0

Rack order: CPU, TM Count, DI, DQ, AI. Memory card, rail, front connectors and load-group wiring require native TIA confirmation.

CPU baseline FS03/FW4.0 is a controlled assumption; delivered hardware state and exact V20 catalog compatibility remain open.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Author: FC01 engineering team	S7-1500 rack and module allocation	File: FC01_revision_e.qet
Date:		Folio: 7/26

E-08 PLC DIGITAL INPUTS I0.0-I1.7

REV E | 8/26

I0.0 -A102/CH0 SAFETY_OK -K100 SPARE/NO FIELD TERM -/-

I0.1 -A102/CH1 GUARD_CLOSED_STATUS -S102 X100:1 C001/1

I0.2 -A102/CH2 AIR_PRESSURE_OK -B100 X100:3 C002/1

I0.3 -A102/CH3 PRODUCT_SUPPLY_OK -B101 X100:6 C003/1

I0.4 -A102/CH4 DI_SPARE_DRIVE_REMOVED_01 SPARE SPARE/NO FIELD TERM -/-

I0.5 -A102/CH5 DI_SPARE_DRIVE_REMOVED_02 SPARE SPARE/NO FIELD TERM -/-

I0.6 -A102/CH6 DI_SPARE_DRIVE_REMOVED_03 SPARE SPARE/NO FIELD TERM -/-

I0.7 -A102/CH7 DI_SPARE_DRIVE_REMOVED_04 SPARE SPARE/NO FIELD TERM -/-

I1.0 -A102/CH8 DI_SPARE_DRIVE_REMOVED_05 SPARE SPARE/NO FIELD TERM -/-

I1.1 -A102/CH9 DI_SPARE_DRIVE_REMOVED_06 SPARE SPARE/NO FIELD TERM -/-

I1.2 -A102/CH10 BOTTLE_AT_NEST_1 -B110 X100:9 C004/1

I1.3 -A102/CH11 BOTTLE_AT_NEST_2 -B111 X100:12 C005/1

I1.4 -A102/CH12 GATE_OPEN_FB -B120 X100:15 C006/1

I1.5 -A102/CH13 GATE_CLOSED_FB -B121 X100:18 C007/1

I1.6 -A102/CH14 CLAMP_RELEASED_FB -B122 X100:21 C008/1

I1.7 -A102/CH15 CLAMP_ENGAGED_FB -B123 X100:24 C009/1

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

E-09 PLC DIGITAL INPUTS I2.0-I3.7 AND SPARES

REV E | 9/26

I2.0 - A102/CH16 FILL_VALVE_1_CLOSED_FB - B130 X101:3 C010/1

I2.1 - A102/CH17 FILL_VALVE_2_CLOSED_FB - B131 X101:6 C011/1

I2.2 - A102/CH18 CAPPER_READY - IF140 X101:9 C012/1

I2.3 - A102/CH19 CAPPER_BUSY - IF140 X101:12 C012/4

I2.4 - A102/CH20 CAPPER_COMPLETE - IF140 X101:13 C012/5

I2.5 - A102/CH21 CAPPER_FAULT - IF140 X101:14 C012/6

I2.6 - A102/CH22 LOCAL_RESET_PB - S100 X101:15 C013/1

I2.7 - A102/CH23 LOCAL_STOP_PB - S101 X101:18 C014/1

I3.0 - A102/CH24 DI_SPARE_01 SPARE SPARE/NO FIELD TERM -/-

I3.1 - A102/CH25 DI_SPARE_02 SPARE SPARE/NO FIELD TERM -/-

I3.2 - A102/CH26 DI_SPARE_03 SPARE SPARE/NO FIELD TERM -/-

I3.3 - A102/CH27 DI_SPARE_04 SPARE SPARE/NO FIELD TERM -/-

I3.4 - A102/CH28 DI_SPARE_05 SPARE SPARE/NO FIELD TERM -/-

I3.5 - A102/CH29 DI_SPARE_06 SPARE SPARE/NO FIELD TERM -/-

I3.6 - A102/CH30 DI_SPARE_07 SPARE SPARE/NO FIELD TERM -/-

I3.7 - A102/CH31 DI_SPARE_08 SPARE SPARE/NO FIELD TERM -/-

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

E-10 PLC DIGITAL OUTPUTS Q0.0-Q1.7

REV E | 10/26

Q0.0 -A103/CH0 DO_SPARE_DRIVE_REMOVED_01 SPARE SPARE/NO FIELD TERM -/-

Q0.1 -A103/CH1 DO_SPARE_DRIVE_REMOVED_02 SPARE SPARE/NO FIELD TERM -/-

Q0.2 -A103/CH2 FILL_VALVE_1_OPEN_CMD -K202 X101:21 C015/1

Q0.3 -A103/CH3 FILL_VALVE_2_OPEN_CMD -K203 X101:23 C016/1

Q0.4 -A103/CH4 GATE_OPEN_CMD -K204 X102:1 C017/1

Q0.5 -A103/CH5 GATE_CLOSE_CMD -K205 X102:3 C017/3

Q0.6 -A103/CH6 CLAMP_ENGAGE_CMD -K206 X102:5 C018/1

Q0.7 -A103/CH7 CLAMP_RELEASE_CMD -K207 X102:7 C018/3

Q1.0 -A103/CH8 CAPPER_REQUEST -K208 X102:9 C012/7

Q1.1 -A103/CH9 STACK_GREEN -K209 X102:11 C019/1

Q1.2 -A103/CH10 STACK_AMBER -K210 X102:13 C019/3

Q1.3 -A103/CH11 STACK_RED -K211 X102:15 C019/5

Q1.4 -A103/CH12 AUDIBLE_ALARM -K212 X102:17 C019/7

Q1.5 -A103/CH13 CAMERA_LIGHT_ENABLE -K213 X102:19 C020/1

Q1.6 -A103/CH14 DO_SPARE_01 SPARE SPARE/NO FIELD TERM -/-

Q1.7 -A103/CH15 DO_SPARE_02 SPARE SPARE/NO FIELD TERM -/-

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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E-11 PLC DIGITAL OUTPUTS Q2.0-Q3.7 AND SPARES

REV E | 11/26

Q2.0 - A103/CH16 DO_SPARE_03 SPARE SPARE/NO FIELD TERM -/-

Q2.1 - A103/CH17 DO_SPARE_04 SPARE SPARE/NO FIELD TERM -/-

Q2.2 - A103/CH18 DO_SPARE_05 SPARE SPARE/NO FIELD TERM -/-

Q2.3 - A103/CH19 DO_SPARE_06 SPARE SPARE/NO FIELD TERM -/-

Q2.4 - A103/CH20 DO_SPARE_07 SPARE SPARE/NO FIELD TERM -/-

Q2.5 - A103/CH21 DO_SPARE_08 SPARE SPARE/NO FIELD TERM -/-

Q2.6 - A103/CH22 DO_SPARE_09 SPARE SPARE/NO FIELD TERM -/-

Q2.7 - A103/CH23 DO_SPARE_10 SPARE SPARE/NO FIELD TERM -/-

Q3.0 - A103/CH24 DO_SPARE_11 SPARE SPARE/NO FIELD TERM -/-

Q3.1 - A103/CH25 DO_SPARE_12 SPARE SPARE/NO FIELD TERM -/-

Q3.2 - A103/CH26 DO_SPARE_13 SPARE SPARE/NO FIELD TERM -/-

Q3.3 - A103/CH27 DO_SPARE_14 SPARE SPARE/NO FIELD TERM -/-

Q3.4 - A103/CH28 DO_SPARE_15 SPARE SPARE/NO FIELD TERM -/-

Q3.5 - A103/CH29 DO_SPARE_16 SPARE SPARE/NO FIELD TERM -/-

Q3.6 - A103/CH30 DO_SPARE_17 SPARE SPARE/NO FIELD TERM -/-

Q3.7 - A103/CH31 DO_SPARE_18 SPARE SPARE/NO FIELD TERM -/-

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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E-12 ANALOG INPUTS AND SHIELD TERMINATION

REV E | 12/26

IW64 -A104/CH0 FLOW_1_MA_RAW -FT100 X102:21 C021/1

IW66 -A104/CH1 FLOW_2_MA_RAW -FT101 X103:1 C022/1

IW68 -A104/CH2 AI_SPARE_01 SPARE SPARE/NO FIELD TERM -/-

IW70 -A104/CH3 AI_SPARE_02 SPARE SPARE/NO FIELD TERM -/-

IW72 -A104/CH4 AI_SPARE_03 SPARE SPARE/NO FIELD TERM -/-

IW74 -A104/CH5 AI_SPARE_04 SPARE SPARE/NO FIELD TERM -/-

IW76 -A104/CH6 AI_SPARE_05 SPARE SPARE/NO FIELD TERM -/-

IW78 -A104/CH7 AI_SPARE_06 SPARE SPARE/NO FIELD TERM -/-

FT100/FT101 are 4-20 mA channels; overall shields/drains terminate at SC100 on the panel end only unless the final EMC study requires otherwise.

AI spare channels are uncommitted. Final MANA grouping, transmitter isolation and fault-range behavior require native configuration.

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E-13 TM COUNT HIGH-SPEED PULSE INPUTS

REV E | 13/26

SENSOR / INSTRUMENT

=FC01+FLD-FT100

Flowmeter pulse channel 1

C021 core 4 to TM1.CH0.A

PLC / MODULE

=FC01+CP01-A101

TM Count 2x24V

CH0 A / CH1 A

SENSOR / INSTRUMENT

=FC01+FLD-FT101

Flowmeter pulse channel 2

C022 core 4 to TM1.CH1.A

TM1.CH0.A-A101/CHCH0 AFLOW_1_PULSE-FT100 X103:5 C021/4

TM1.CH1.A-A101/CHCH1 AFLOW_2_PULSE-FT101 X103:7 C022/4

Pulse and 4-20 mA signals are independent diagnostics from each flowmeter; neither substitutes for the other.

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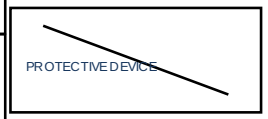
Author: FC01 engineering team	TM Count high-speed pulse inputs	File: FC01_revision_e.qet
Date:		Folio: 13/26

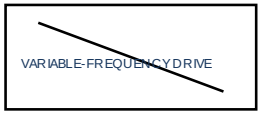
E-14 CONVEYOR G120C POWER AND PROFINET CONTROL

REV E | 14/26



=FC01+CP01-QF100

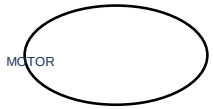
Provisional drive branch protection



=FC01+CP01-U100

G120C PN conveyor drive

Standard Telegram 1; IW/QW256-259



=FC01+FLD-M100

Conveyor motor

Nameplate and duty open

=FC01+CP01-PE100

Motor PE/bonding

PLC controls STW1 and speed reference; drive returns ZSW1 and actual speed. No hardwired ordinary run/ready path is claimed.

External STO is conceptual safety scope and is not controlled or credited by the standard PLC.

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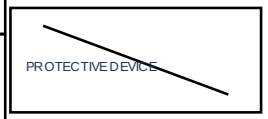
Author: FC01 engineering team	Conveyor G120C power and PROFINET control	File: FC01_revision_e.qet
Date:		Folio: 14/26

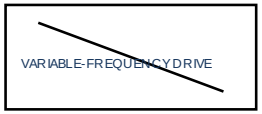
E-15 PUMP G120C POWER AND PROFINET CONTROL

REV E | 15/26



=FC01+CP01-QF100

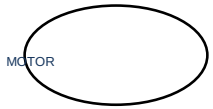
Provisional drive branch protection



=FC01+CP01-U101

G120C PN dosing-pump drive

Standard Telegram 1; IW/QW 260-263



=FC01+FLD-M101

Pump motor

Nameplate and hydraulic duty open

=FC01+CP01-PE100

Motor PE/bonding

Pump speed, ramps, current limits and overload duty remain provisional until motor and pump nameplates and process duty are supplied.

External STO is conceptual safety scope and is not controlled or credited by the standard PLC.

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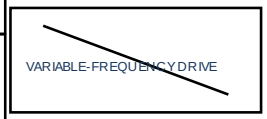
Author: FC01 engineering team	Pump G120C power and PROFINET control	File: FC01_revision_e.qet
Date:		Folio: 15/26

E-16 MOTOR CABLES, PE AND SHIELD TERMINATION

REV E | 16/26

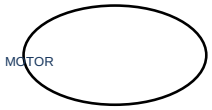


=FC01+CP01-U100

Conveyor drive

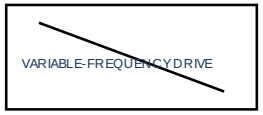
=FC01+CP01-SC100

360-degree shield-clamp rail



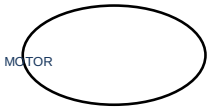
=FC01+FLD-M100

Conveyor motor



=FC01+CP01-U101

Pump drive



=FC01+FLD-M101

Pump motor

W901 denotes the controlled requirement for shielded VFD motor cables and EMC termination kits; exact cable identifiers and routes remain open.

Maintain short, low-impedance shield terminations and continuous PE. Final routing, ampacity and EMC checks require installed geometry.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Author: FC01 engineering team	Motor cables, PE and shield termination	File: FC01_revision_e.qet
Date:		Folio: 16/26

E-17 PROFINET AND OT NETWORK ARCHITECTURE

REV E | 17/26

NETWORK NODE

=FC01+CP01-FW100

SCALANCE S615 boundary

NETWORK NODE

=FC01+CP01-SW100

SCALANCE XC208 managed switch

PLC / MODULE

=FC01+CP01-A100

PLC / OPC UA server

NETWORK NODE

=FC01+CP01-H100

MTP700 Unified HMI

NETWORK NODE

=FC01+CP01-PC200

NVIDIA OPC UA client

PLC-A100 192.168.10.10 VLAN 10 XC208 P1 CONTROL PROFINET controller / OPC UA server

HMI-H100 192.168.10.20 VLAN 10 XC208 P2 CONTROL Unified operator panel

VFD-U100 192.168.10.31 VLAN 10 XC208 P3 CONTROL Conveyor G120C PN

VFD-U101 192.168.10.32 VLAN 10 XC208 P4 CONTROL Pump G120C PN

VISION-PC200 192.168.20.10 VLAN 20 XC208 P5 QUALITY OPC UA client + inference

ENG-SERVICE 192.168.99.0/24 VLAN 99 XC208 P6 ENGINEERING Temporary service

FW-S615 Zone gateways VLAN 10/20/99 XC208 P7 trunk BOUNDARY Router/firewall

U100 and U101 are PROFINET nodes on the CONTROL VLAN; PC200 remains in QUALITY and never owns hazardous motion commands.

Firewall rules, certificates, NTP, logging and engineering access require site cybersecurity approval.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	PROFINET and OT network architecture	File: FC01_revision_e.qet
Date:		Folio: 17/26

E-18 HMI AND NVIDIA INSPECTION INTERFACE

REV E | 18/26

NETWORK NODE

=FC01+CP01-H100

MTP700 Unified Comfort

PLC / MODULE

=FC01+CP01-A100

Sequence and interlock authority

NETWORK NODE

=FC01+CP01-PC200

Non-safety quality inspection

SENSOR / INSTRUMENT

=FC01+FLD-CAM200

Industrial camera

ACTUATOR / RELAY

=FC01+FLD-LT200

Controlled vision lighting

OPC UA transaction uses nonzero session/inspection identity, held request handshake, immutable result publication and exact acknowledgement.

Missing, stale, duplicate, contradictory, uncertain, model-mismatched or timed-out AI data produces deterministic HOLD or reject disposition.

NVIDIA cannot command physical outputs, drives, sequence transitions or bypass PLC interlocks.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	HMI and NVIDIA inspection interface	File: FC01_revision_e.qet
Date:		Folio: 18/26

E-19 SOLENOID VALVES AND INTERPOSING RELAYS

REV E | 19/26

PLC / MODULE

=FC01+CP01-A103

DQ 32x24V DC

ACTUATOR / RELAY

=FC01+CP01-K202..-K213

Interposing relay family

ACTUATOR / RELAY

=FC01+FLD-YV100

Gate solenoid

C017

ACTUATOR / RELAY

=FC01+FLD-YV101

Clamp solenoid

C018

ACTUATOR / RELAY

=FC01+FLD-YV102

Fill valve 1 fail closed

C015

ACTUATOR / RELAY

=FC01+FLD-YV103

Fill valve 2 fail closed

C016

K202-K213 relay coils require D202-D213 suppression. Gate/clamp opposing commands are interlocked in PLC logic.

Fill valves are fail closed; de-energized state is closed. Final coil current and protection coordination remain open.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

E-20 FIELD SENSORS, FLOW AND LEVEL INSTRUMENTATION

REV E | 20/26

SENSOR / INSTRUMENT

=FC01+FLD-B100

Air pressure switch

SENSOR / INSTRUMENT

=FC01+FLD-B101

Product supply permissive

SENSOR / INSTRUMENT

=FC01+FLD-B110

Bottle nest 1

SENSOR / INSTRUMENT

=FC01+FLD-B111

Bottle nest 2

SENSOR / INSTRUMENT

=FC01+FLD-FT100

Flow channel 1 pulse + 4-20 mA

SENSOR / INSTRUMENT

=FC01+FLD-FT101

Flow channel 2 pulse + 4-20 mA

SENSOR / INSTRUMENT

=FC01+FLD-B120

Gate open proof

SENSOR / INSTRUMENT

=FC01+FLD-B121

Gate closed proof

SENSOR / INSTRUMENT

=FC01+FLD-B122

Clamp released proof

SENSOR / INSTRUMENT

=FC01+FLD-B123

Clamp engaged proof

B130/B131 provide fill-valve closed proof; LT200 is vision lighting, not a process-level transmitter.

All ordinary sensing is non-safety unless separately designed and validated by a qualified machinery-safety engineer.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Field sensors, flow and level instrumentation	File: FC01_revision_e.qet
Date:		Folio: 20/26

E-21 STACK LIGHT, LOCAL STATIONS AND CAPPER INTERFACE

REV E | 21/26

DEVICE

=FC01+FLD-S100

Local reset pushbutton

C013

DEVICE

=FC01+FLD-S101

Local controlled-stop pushbutton

C014

ACTUATOR / RELAY

=FC01+FLD-H101

Three-color stack light and sounder

C019

TERMINAL / SHIELD

=FC01+FLD-IF140

Capper ready/busy/complete/fault/request

C012

PLC / MODULE

=FC01+CP01-A102

DI capper/local feedback

PLC / MODULE

=FC01+CP01-A103

DO request/indication

Capper BUSY/COMPLETE is accepted only after a correlated request. Unsolicited status is a deterministic interface fault.

Local stop is a controlled process stop, not an emergency-stop or protective device.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Stack light, local stations and capper interface	File: FC01_revision_e.qet
Date:		Folio: 21/26

E-22 TERMINAL STRIPS X100 AND X101

X100:1 GUARD_CLOSED_STATUS C001 /1 I0.1 24 VDC switched W0003

X100:2 GUARD_CLOSED_STATUS C001 /2 I0.1 +24VDC fused W0004

X100:3 AIR_PRESSURE_OK C002 /1 I0.2 24 VDC switched W0005

X100:4 AIR_PRESSURE_OK C002 /2 I0.2 +24VDC fused W0006

X100:5 AIR_PRESSURE_OK C002 /3 I0.2 0V W0007

X100:6 PRODUCT_SUPPLY_OK C003 /1 I0.3 24 VDC switched W0008

X100:7 PRODUCT_SUPPLY_OK C003 /2 I0.3 +24VDC fused W0009

X100:8 PRODUCT_SUPPLY_OK C003 /3 I0.3 0V W0010

X100:9 BOTTLE_AT_NEST_1 C004 /1 I1.2 24 VDC switched W0011

X100:10 BOTTLE_AT_NEST_1 C004 /2 I1.2 +24VDC fused W0012

X100:11 BOTTLE_AT_NEST_1 C004 /3 I1.2 0V W0013

X100:12 BOTTLE_AT_NEST_2 C005 /1 I1.3 24 VDC switched W0014

X100:13 BOTTLE_AT_NEST_2 C005 /2 I1.3 +24VDC fused W0015

X100:14 BOTTLE_AT_NEST_2 C005 /3 I1.3 0V W0016

X100:15 GATE_OPEN_FB C006 /1 I1.4 24 VDC switched W0017

X100:16 GATE_OPEN_FB C006 /2 I1.4 +24VDC fused W0018

X100:17 GATE_OPEN_FB C006 /3 I1.4 0V W0019

X100:18 GATE_CLOSED_FB C007 /1 I1.5 24 VDC switched W0020

X100:19 GATE_CLOSED_FB C007 /2 I1.5 +24VDC fused W0021

X100:20 GATE_CLOSED_FB C007 /3 I1.5 0V W0022

X100:21 CLAMP_RELEASED_FB C008 /1 I1.6 24 VDC switched W0023

X100:22 CLAMP_RELEASED_FB C008 /2 I1.6 +24VDC fused W0024

X100:23 CLAMP_RELEASED_FB C008 /3 I1.6 0V W0025

X100:24 CLAMP_ENGAGED_FB C009 /1 I1.7 24 VDC switched W0026

X101:1 CLAMP_ENGAGED_FB C009 /2 I1.7 +24VDC fused W0027

X101:2 CLAMP_ENGAGED_FB C009 /3 I1.7 0V W0028

X101:3 FILL_VALVE_1_CLOSED_FB C010 /1 I2.0 24 VDC switched W0029

X101:4 FILL_VALVE_1_CLOSED_FB C010 /2 I2.0 +24VDC fused W0030

X101:5 FILL_VALVE_1_CLOSED_FB C010 /3 I2.0 0V W0031

X101:6 FILL_VALVE_2_CLOSED_FB C011 /1 I2.1 24 VDC switched W0032

X101:7 FILL_VALVE_2_CLOSED_FB C011 /2 I2.1 +24VDC fused W0033

X101:8 FILL_VALVE_2_CLOSED_FB C011 /3 I2.1 0V W0034

X101:9 CAPPER_READY C012 /1 I2.2 24 VDC switched W0035

X101:10 CAPPER_READY C012 /2 I2.2 +24VDC fused W0036

X101:11 CAPPER_READY C012 /3 I2.2 0V W0037

X101:12 CAPPER_BUSY C012 /4 I2.3 24 VDC switched W0038

X101:13 CAPPER_COMPLETE C012 /5 I2.4 24 VDC switched W0039

X101:14 CAPPER_FAULT C012 /6 I2.5 24 VDC switched W0040

X101:15 LOCAL_RESET_PB C013 /1 I2.6 24 VDC switched W0041

X101:16 LOCAL_RESET_PB C013 /2 I2.6 +24VDC fused W0042

X101:17 LOCAL_RESET_PB C013 /3 I2.6 0V W0043

X101:18 LOCAL_STOP_PB C014 /1 I2.7 24 VDC switched W0044

X101:19 LOCAL_STOP_PB C014 /2 I2.7 +24VDC fused W0045

X101:20 LOCAL_STOP_PB C014 /3 I2.7 0V W0046

X101:21 FILL_VALVE_1_OPEN_CMD C015 /1 Q0.2 +24VDC switched W0050

X101:22 FILL_VALVE_1_OPEN_CMD C015 /2 Q0.2 0V W0051

X101:23 FILL_VALVE_2_OPEN_CMD C016 /1 Q0.3 +24VDC switched W0055

X101:24 FILL_VALVE_2_OPEN_CMD C016 /2 Q0.3 0V W0056

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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E-23 TERMINAL STRIPS X102/X103, POWER, COMMON AND SHIELDS

REV E | 23/26

X102:1 GATE_OPEN_CMD C017 /1 Q0.4 +24VDC switched W0060

X102:2 GATE_OPEN_CMD C017 /2 Q0.4 0V W0061

X102:3 GATE_CLOSE_CMD C017 /3 Q0.5 +24VDC switched W0065

X102:4 GATE_CLOSE_CMD C017 /4 Q0.5 0V W0066

X102:5 CLAMP_ENGAGE_CMD C018 /1 Q0.6 +24VDC switched W0070

X102:6 CLAMP_ENGAGE_CMD C018 /2 Q0.6 0V W0071

X102:7 CLAMP_RELEASE_CMD C018 /3 Q0.7 +24VDC switched W0075

X102:8 CLAMP_RELEASE_CMD C018 /4 Q0.7 0V W0076

X102:9 CAPPER_REQUEST C012 /7 Q1.0 +24VDC switched W0080

X102:10 CAPPER_REQUEST C012 /8 Q1.0 0V W0081

X102:11 STACK_GREEN C019 /1 Q1.1 +24VDC switched W0085

X102:12 STACK_GREEN C019 /2 Q1.1 0V W0086

X102:13 STACK_AMBER C019 /3 Q1.2 +24VDC switched W0090

X102:14 STACK_AMBER C019 /4 Q1.2 0V W0091

X102:15 STACK_RED C019 /5 Q1.3 +24VDC switched W0095

X102:16 STACK_RED C019 /6 Q1.3 0V W0096

X102:17 AUDIBLE_ALARM C019 /7 Q1.4 +24VDC switched W0100

X102:18 AUDIBLE_ALARM C019 /8 Q1.4 0V W0101

X102:19 CAMERA_LIGHT_ENABLE C020 /1 Q1.5 +24VDC switched W0105

X102:20 CAMERA_LIGHT_ENABLE C020 /2 Q1.5 0V W0106

X102:21 FLOW_1_MA_RAW C021 /1 IW64 4-20mA W0107

X102:22 FLOW_1_MA_RAW C021 /2 IW64 +24VDC fused W0108

X102:23 FLOW_1_MA_RAW C021 /3 IW64 0V analog W0109

X102:24 FLOW_1_MA_RAW C021 /SH IW64 FE W0110

X103:1 FLOW_2_MA_RAW C022 /1 IW66 4-20mA W0111

X103:2 FLOW_2_MA_RAW C022 /2 IW66 +24VDC fused W0112

X103:3 FLOW_2_MA_RAW C022 /3 IW66 0V analog W0113

X103:4 FLOW_2_MA_RAW C022 /SH IW66 FE W0114

X103:5 FLOW_1_PULSE C021 /4 TM1.CH0.A 24 V pulse W0115

X103:6 FLOW_1_PULSE C021 /5 TM1.CH0.A 0V W0116

X103:7 FLOW_2_PULSE C022 /4 TM1.CH1.A 24 V pulse W0117

X103:8 FLOW_2_PULSE C022 /5 TM1.CH1.A 0V W0118

X24:+24V X24:+24V - /- - +24VDC

X0:0V X0:0V - /- - 0V

X0:MANA X0:MANA - /- - 0V analog

SC100:1 SC100:1 - /- - FE

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Terminal strips X102/X103, power, common and shields	File: FC01_revision_e.qet
Date:		Folio: 23/26

E-24 CABLE SCHEDULE AND EXTERNAL FIELD CONNECTIONS

REV E | 24/26

C001 =FC01+CP01 panel terminals -> =FC01+FLD-S102 Industrial control cores 2/4 None scheduled 24 VDC control

C002 =FC01+CP01 panel terminals -> =FC01+FLD-B100 Industrial control cores 3/4 None scheduled 24 VDC control

C003 =FC01+CP01 panel terminals -> =FC01+FLD-B101 Industrial control cores 3/4 None scheduled 24 VDC control

C004 =FC01+CP01 panel terminals -> =FC01+FLD-B110 Industrial control cores 3/4 None scheduled 24 VDC control

C005 =FC01+CP01 panel terminals -> =FC01+FLD-B111 Industrial control cores 3/4 None scheduled 24 VDC control

C006 =FC01+CP01 panel terminals -> =FC01+FLD-B120 Industrial control cores 3/4 None scheduled 24 VDC control

C007 =FC01+CP01 panel terminals -> =FC01+FLD-B121 Industrial control cores 3/4 None scheduled 24 VDC control

C008 =FC01+CP01 panel terminals -> =FC01+FLD-B122 Industrial control cores 3/4 None scheduled 24 VDC control

C009 =FC01+CP01 panel terminals -> =FC01+FLD-B123 Industrial control cores 3/4 None scheduled 24 VDC control

C010 =FC01+CP01 panel terminals -> =FC01+FLD-B130 Industrial control cores 3/4 None scheduled 24 VDC control

C011 =FC01+CP01 panel terminals -> =FC01+FLD-B131 Industrial control cores 3/4 None scheduled 24 VDC control

C012 =FC01+CP01 panel terminals -> =FC01+FLD-IF140 Industrial control cores 8/12 None scheduled 24 VDC control

C013 =FC01+CP01 panel terminals -> =FC01+FLD-S100 Industrial control cores 3/4 None scheduled 24 VDC control

C014 =FC01+CP01 panel terminals -> =FC01+FLD-S101 Industrial control cores 3/4 None scheduled 24 VDC control

C015 =FC01+CP01 panel terminals -> =FC01+FLD-YV102 Industrial control cores 2/4 None scheduled 24 VDC control

C016 =FC01+CP01 panel terminals -> =FC01+FLD-YV103 Industrial control cores 2/4 None scheduled 24 VDC control

C017 =FC01+CP01 panel terminals -> =FC01+FLD-YV100 Industrial control cores 4/8 None scheduled 24 VDC control

C018 =FC01+CP01 panel terminals -> =FC01+FLD-YV101 Industrial control cores 4/8 None scheduled 24 VDC control

C019 =FC01+CP01 panel terminals -> =FC01+FLD-H101 Industrial control cores 8/12 None scheduled 24 VDC control

C020 =FC01+CP01 panel terminals -> =FC01+FLD-LT200 Industrial control cores 2/4 None scheduled 24 VDC control

C021 =FC01+CP01 panel terminals -> =FC01+FLD-FT100 Shielded instrumentation cores 5/8 Separate overall shield/drain to SC100; not counted as an insulated core Analog/instrument

C022 =FC01+CP01 panel terminals -> =FC01+FLD-FT101 Shielded instrumentation cores 5/8 Separate overall shield/drain to SC100; not counted as an insulated core Analog/instrument

Cable/core identifiers on this folio are copied from the controlled schedule; conductor sizes remain blank pending site and installation data.

C021/C022 instrumentation shields terminate at SC100. Power and VFD motor cables remain open schedule inputs.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Cable schedule and external field connections	File: FC01_revision_e.qet
Date:		Folio: 24/26

E-25 CONCEPTUAL SAFETY INTERFACE AND BOUNDARIES

REV E | 25/26

CONCEPTUAL SAFETY INTERFACE

=FC01+FLD-S102

Guard-status monitoring mirror

CONCEPTUAL SAFETY INTERFACE

=FC01+CP01-K100

External safety-interface monitoring contact

PLC / MODULE

=FC01+CP01-A102

Ordinary PLC status monitoring only

VARIABLE-FREQUENCY DRIVE

=FC01+CP01-U100

External STO conceptual interface

VARIABLE-FREQUENCY DRIVE

=FC01+CP01-U101

External STO conceptual interface

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGUL

Standard PLC, HMI and NVIDIA functions receive status only and receive no safety credit.

OPEN: hazard analysis, protective-device selection, reset/restart design, PL/SIL/category determination and validation.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

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Author: FC01 engineering team	Conceptual safety interface and boundaries	File: FC01_revision_e.qet
Date:		Folio: 25/26

E-26 SPARE CAPACITY, BOM COVERAGE AND OPEN INPUTS

REV E | 26/26

=FC01+CP01-A100 SIMATIC S7-1500 CPU 1511-1 PN Selected baseline

=FC01+CP01-A101 TM Count 2x24V Selected baseline

=FC01+CP01-A102 DI 32x24V DC HF Selected baseline

=FC01+CP01-A103 DQ 32x24V DC/0.5A HF Selected baseline

=FC01+CP01-A104 AI 8xU/I HF Selected baseline

=FC01+CP01-H100 MTP700 Unified Comfort Selected baseline

=FC01+CP01-U100 SINAMICS G120C PN 0.75 kW unfiltered Provisional

=FC01+CP01-U101 SINAMICS G120C PN 0.75 kW unfiltered Provisional

=FC01+CP01-SW100 SCALANCE XC208 managed Selected network baseline

=FC01+CP01-FW100 SCALANCE S615 EEC Selected network baseline

=FC01+CP01-PC200 Jetson Orin NX 16GB on industrial carrier Architecture only

=FC01+CP01-MC100 SIMATIC memory card; capacity to be selected after compiled project sizing TBD

=FC01+CP01-RA100 S7-1500 mounting rail and fixing hardware TBD

=FC01+CP01-FC100 Front connector for DI module TBD

=FC01+CP01-FC101 Front connector for DQ module TBD

=FC01+CP01-FC102 Front connector for AI module TBD

=FC01+CP01-FC103 Front connector for TM Count module TBD

=FC01+CP01-QS100 Main door-coupled isolator TBD

=FC01+CP01-QF100 Main protective device TBD

=FC01+CP01-PS100 24 VDC 20 A power supply Provisional

=FC01+CP01-QF110 24 VDC electronic branch protection TBD

=FC01+CP01-K100 Main control enable contactor / external safety interface with dry monitoring contact TBD

=FC01+CP01-K202..-K213 Interposing relays with bases and status indication TBD

=FC01+CP01-D202..-D213 Coil suppression modules TBD

=FC01+CP01-X100..-X199 DIN-rail terminal blocks, disconnect/test where required Provisional quantity

=FC01+CP01-PE100 Copper protective-earth bar and bonding hardware TBD

=FC01+CP01-SC100 Shield-clamp rail and spring clamps Provisional quantity

=FC01+CP01-ENC100 IP-rated enclosure with mounting plate TBD

=FC01+CP01-WD100 Slotted wiring duct set TBD

=FC01+CP01-DR100 35 mm DIN rail set TBD

=FC01+FLD-GL100 Cable glands and EMC glands TBD

=FC01+CP01-FAN100 Filter fan / cooling solution TBD

=FC01+FLD-M100 Conveyor motor TBD

=FC01+FLD-M101 Pump motor TBD

=FC01+FLD-YV100 Gate valve/cylinder solenoid TBD

=FC01+FLD-YV101 Clamp valve/cylinder solenoid TBD

=FC01+FLD-YV102 Fill valve channel 1, fail closed TBD

=FC01+FLD-YV103 Fill valve channel 2, fail closed TBD

=FC01+FLD-FT100 Pulse plus 4-20 mA flowmeter channel 1 TBD

=FC01+FLD-FT101 Pulse plus 4-20 mA flowmeter channel 2 TBD

=FC01+FLD-S100 Local reset pushbutton TBD

=FC01+FLD-S101 Local controlled-stop pushbutton TBD

=FC01+FLD-S102 Guard-status dry-contact monitoring interface TBD

=FC01+FLD-H101 Three-color stack light with sounder TBD

=FC01+FLD-IF140 Capper interface terminals/relays TBD

=FC01+FLD-CAM200 Industrial camera TBD

=FC01+FLD-LENS200 C-mount lens TBD

=FC01+FLD-LT200 Machine-vision backlight/front-light assembly TBD

=FC01+FLD-CB200 Industrial Jetson carrier TBD

=FC01+FLD-W900 Industrial Ethernet patch cables/connectors Provisional quantity

=FC01+FLD-W901 Shielded VFD motor cables and EMC termination kits TBD

=FC01+FLD-CONS100 Markers, ferrules, lugs, ties and installation consumables TBD

=FC01+FLD-B100 Pneumatic pressure switch TBD

=FC01+FLD-B101 Product supply permissive TBD

=FC01+FLD-B110 Bottle 1 position sensor TBD

=FC01+FLD-B111 Bottle 2 position sensor TBD

=FC01+FLD-B120 Gate open feedback TBD

=FC01+FLD-B121 Gate closed feedback TBD

=FC01+FLD-B122 Clamp released feedback TBD

=FC01+FLD-B123 Clamp engaged feedback TBD

=FC01+FLD-B130 Valve 1 closed proof TBD

=FC01+FLD-B131 Valve 2 closed proof TBD

Controlled spare capacity: 8 DI, 18 DO and 6 AI allocated spares; spare terminals/cores require final allocation and labeling.

OPEN: supply/fault current, motor and pump data, cable routes, conductor sizing, thermal/EMC study, exact PC200 envelope and construction review.

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Author: FC01 engineering team	Spare capacity, BOM coverage and open inputs	File: FC01_revision_e.qet
Date:		Folio: 26/26